Chapter III.5, Week 12

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5.1. Let X be a projective scheme over a field k, and let $\mathscr{F}$ be a coherent sheaf on X. We define the *Euler characteristic* of $\mathscr{F}$ by

$$\chi(\mathscr{F}) = \sum_i (-1)^i \dim_k H^i(X, \mathscr{F}).$$

If

$$0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$$

is a short exact sequence of coherent sheaves on X, show that $\chi(\mathscr{F}) = \chi(\mathscr{F}')+\chi(\mathscr{F}'')$.

Proof. First, for an exact sequence of vector spaces $0 \to V_0 \xrightarrow{f_0} \cdots \xrightarrow{f_{n-1}} V_n \xrightarrow{f_n} 0$, consider $\sum_i (-1)^i \dim V_i$. We have

$$\begin{aligned}\sum_{i=0}^{n} (-1)^i \dim V_i &= \sum_{i=1}^{n} (-1)^i \dim \ker f_i + \sum_{i=0}^{n-1} (-1)^i \dim \operatorname{im} f_i \\ &= \sum_{i=1}^{n} (-1)^i \dim \ker f_i + \sum_{i=0}^{n-1} (-1)^i \dim \ker f_{i+1} \\ &= 0.\end{aligned}$$

Thus, the long exact sequence associated to $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ gives us

$$\chi(\mathscr{F}') - \chi(\mathscr{F}) + \chi(\mathscr{F}'') = 0,$$

as desired. □

5.2. (a) Let X be a projective scheme over a field k, let $\mathscr{O}_X(1)$ be a very ample invertible sheaf on X over k, and let $\mathscr{F}$ be a coherent sheaf on X. Show that there is a polynomial $P(z) \in \mathbb{Q}[z]$, such that $\chi(\mathscr{F}(n)) = P(n)$ for all $n \in \mathbb{Z}$. We call P the *Hilbert polynomial* of $\mathscr{F}$ with respect to the sheaf $\mathscr{O}_X(1)$.
[*Hint*: Use induction on $\dim \operatorname{Supp} \mathscr{F}$, general properties of numerical polynomials (I, 7.3), and suitable exact sequences

$$0 \to \mathscr{R} \to \mathscr{F}(-1) \to \mathscr{F} \to \mathscr{D} \to 0.]$$

(b) Now let $X = \mathbb{P}^r_k$, and let $M = \Gamma_*(\mathscr{F})$, considered as a graded $S = k[x_0, \ldots, x_r]$-module. Use (5.2) to show that the Hilbert polynomial of $\mathscr{F}$ just defined is the same as the Hilbert polynomial of M defined in (I, 7).

Proof. (a)

(b) Let $P(z)$ be the Hilbert poylnomial of $\mathscr{F}$ defined in (a). It suffices to show that $P(n) = \dim M_n = \dim_k H^0(X, \mathscr{F}(n))$ for $n \gg 0$. However, (5.2) gives us

$$\begin{aligned} P(n) &= \chi(\mathscr{F}(n)) \\ &= \sum_{i=0}^{n} (-1)^i H^i(X, \mathscr{F}(n)) \\ &= H^0(X, \mathscr{F}(n)) \end{aligned}$$

for $n \gg 0$, so we're done. □

5.3. *Arithmetic Genus.* Let X be a projective scheme of dimension r over a field k. We define the *arithmetic genus* p_a of X by

$$p_a(X) = (-1)^r(\chi(\mathscr{O}_X) - 1).$$

Note that it depends only on X, not on any projective embedding.

(a) If X is integral, and k algebraically closed, show that $H^0(X, \mathscr{O}_X) \cong k$, so that

$$p_a(X) = \sum_{i=0}^{r-1} (-1)^i \dim_k H^{r-i}(X, \mathscr{O}_X).$$

In particular, if X is a curve, we have

$$p_a(X) = \dim_k H^1(X, \mathscr{O}_X).$$

[*Hint*: Use (I, 3.4).]

(b) If X is a closed subvariety of $\mathbb{P}^r_k$, show that this $p_a(X)$ coincidese with the one defined in (I, Ex. 7.2), which apparently depended on the projective embedding.

(c) If X is a nonsingular projective curve over an algebraically closed field k, show that $p_a(X)$ is in fact a *birational* invariant. Conclude that a nonsingular plane curve of degree $d \geqslant 3$ is not rational. (This gives another proof of (II, 8.20.3) where we used the geometric genus.)

Proof. (a) We have that $H^0(X, \mathscr{O}_X) \cong k$ by (I, 3.4). Then

$$\begin{aligned} p_a(X) &= (-1)^r(\chi(\mathscr{O}_X) - 1) \\ &= (-1)^r(\chi(\mathscr{O}_X) - \dim_k H^0(X, \mathscr{O}_X)) \\ &= (-1)^r(\sum_{i=1}^{r} (-1)^i \dim_k H^i(X, \mathscr{O}_X)) \\ &= \sum_{i=1}^{r} (-1)^{r-i} \dim_k H^i(X, \mathscr{O}_X) \\ &= \sum_{i=0}^{r-1} (-1)^i \dim_k H^i(X, \mathscr{O}_X). \end{aligned}$$

If X is a curve, then $r = 1$, so the result holds.

(b)

(c) □

5.4. Recall from (II, Ex. 6.10) the definition of the Grothendieck group $K(X)$ of a noetherian scheme X.

(a) Let X be a projective scheme over a field k, and let $\mathscr{O}_X(a)$ be a very ample invertible sheaf on X. Show that there is a (unique) additive homomorphism

$$P : K(X) \to \mathbb{Q}[z]$$

such that for each coherent sheaf $\mathscr{F}$ on X, $P(\gamma(\mathscr{F}))$ is the Hilbert polynomial of $\mathscr{F}$ (Ex. 5.2).

(b) Now let $X = \mathbb{P}^r_k$. For each $i = 0, 1, \ldots, r$, let L_i be a linear space of dimension i in X. Then show that

(i) $K(X)$ is the free abelian group generated by $\{\gamma(\mathscr{O}_{L_i}) \mid i = 0, \ldots, r\}$, and

(ii) the map $P : K(X) \to \mathbb{Q}[z]$ is injective.

[*Hint*: Show that (i) $\Longrightarrow$ (ii). Then prove (i) and (ii) simultaneously, by induction on r, using (II, Ex. 6.10c).]

Proof. (a) The obvious map on the free abelian group generated by all coherent sheaves of X descends to $K(X)$ by (Ex. 5.1) above.

(b) If (i) holds, then P is injective because $P(\gamma(\mathscr{O}_{L_i}))$ □

5.7. Let X, Y be proper schemes over a noetherian ring A. We denote by $\mathscr{L}$ an invertible sheaf.

(a) If $\mathscr{L}$ is ample on X, and Y is any closed subscheme of X, then $i^*\mathscr{L}$ is ample on Y, where $i : Y \to X$ is the inclusion.

(b) $\mathscr{L}$ is ample on X if and only if $\mathscr{L}_{\text{red}} = \mathscr{L} \otimes \mathscr{O}_{X_{\text{red}}}$ is ample on X_{red}.

(c) Suppose X is reduced. Then $\mathscr{L}$ is ample on X if and only if $\mathscr{L} \otimes \mathscr{O}_{X_i}$ is ample on X_i for each irreducible component X_i of X.

(d) Let $f : X \to Y$ be a finite surjective morphism, and let $\mathscr{L}$ be an invertible sheaf on Y. Then $\mathscr{L}$ is ample on Y if and only if $f^*\mathscr{L}$ is ample on X.

[*Hints*: Use (5.3) and compare (Ex. 3.1, Ex. 3.2, Ex. 4.1, Ex. 4.2).]

Proof. (a) Let $\mathscr{F}$ be a coherent sheaf on Y. i^* is a left adjoint functor and tensor products are direct limits, so $(i^*\mathscr{L})^n = i^*(\mathscr{L}^n)$. Thus, we may omit the parentheses without ambiguity. By (II, Ex. 5.1d), we have

$$i_*(\mathscr{F} \otimes_{\mathscr{O}_Y} i^*\mathscr{L}^n) \cong i_*\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^n.$$

X, Y are noetherian and i is a closed immersion, so $i_*\mathscr{F}$ is coherent by (II, Ex. 5.5). Then (5.3) gives us $n_0 > 0$ such that, for all $n > 0$ and $p > 0$,

$$H^p(Y, \mathscr{F} \otimes i^*\mathscr{L}^n) \cong H^p(X, i_*\mathscr{F} \otimes \mathscr{L}^n) \cong 0$$

by (2.10). By (5.3) once more, we conclude that $i^*\mathscr{L}$ is ample.

(b) We begin by noting that the immersion $i : X_{\text{red}} \rightarrowtail X$ is a bijection, so $i^*\mathscr{L} = \mathscr{L} \otimes \mathscr{O}_X = \mathscr{L}_{\text{red}}$. If $\mathscr{L}$ is ample on X, then $\mathscr{L}_{\text{red}}$ is ample on X_{red} by (a). Suppose then that $\mathscr{L}_{\text{red}}$ is ample on X_{red} and fix a coherent sheaf $\mathscr{F}$ on X. As in (3.1), noetherianity of X gives a finite filtration $\mathscr{F} \supset \mathscr{N}\mathscr{F} \supset \cdots \supset \mathscr{N}^n\mathscr{F} = 0$, and, thus, the exact sequences

$$0 \to \mathscr{N}^{k+1}\mathscr{F} \to \mathscr{N}^k\mathscr{F} \to \frac{\mathscr{N}^k\mathscr{F}}{\mathscr{N}^{k+1}\mathscr{F}} \to 0$$

for $0 \leqslant k < n$.

Note that the quotient sheaves are coherent by coherence of $\mathscr{N}^k\mathscr{F}$ for all $k \geqslant 0$. Then

$$\mathscr{N} \subset \operatorname{Ann}\left(\frac{\mathscr{N}^k\mathscr{F}}{\mathscr{N}^{k+1}\mathscr{F}}\right),$$

supplies us with sheaves $\mathscr{G}_k$ on X_{red} such that $i_*\mathscr{G}_k$. In fact, the $\mathscr{G}_k$ are coherent. This is because $\operatorname{im}\overline{f} = \operatorname{im}(\overline{f}\circ\pi) = \operatorname{im} f$ whenever π is surjective, so in particular when $\pi : A \to A/\mathfrak{a}$ is the projection. Then (5.3) supplies $n_0 > 0$ such that, for all $n \geqslant n_0$ and $p > 0$, $H^p(X_{\mathrm{red}}, \mathscr{G}_k \otimes \mathscr{L}^n) \cong 0$.

Applying the projection formula (II, 5.1) gives us

$$H^\bullet(X, \frac{\mathscr{N}^k\mathscr{F}}{\mathscr{N}^{k+1}\mathscr{F}} \otimes \mathscr{L}^n) \cong H^\bullet(X_{\mathrm{red}}, \mathscr{G}_k \otimes \mathscr{L}^n).$$

Tensoring our original exact sequence with $\mathscr{L}^n$ for $n \geqslant n_0$ and taking the long exact sequence gives us exactness of

$$H^p(X, \mathscr{N}^{k+1}\mathscr{F} \otimes \mathscr{L}^n) \to H^p(X, \mathscr{N}^k\mathscr{F} \otimes \mathscr{L}^n) \to H^p(X_{\mathrm{red}}, \mathscr{G}_k \otimes \mathscr{L}^n) = 0$$

for $p > 0$. $\mathscr{N}^n\mathscr{F} = 0$, so it follows by induction on $n - k$ that

$$H^p(X, \mathscr{F} \otimes \mathscr{L}^n) = 0$$

for all $p > 0$, whence $\mathscr{L}$ is ample by (5.3).

(c) $\mathscr{L}_i := \mathscr{L} \otimes X_i$ is slightly abusive notation for the inverse image of the inclusion $X_i \hookrightarrow X$, where X_i is given the reduced induced subscheme structure. Thus, one direction is a consequence of (a). It suffices to show that $\mathscr{L}$ is ample if $\mathscr{L}_i$ is ample for each irreducible component X_i of X.

Let $X = X_1 \cup \cdots \cup X_r$ be an irreducible decomposition of X, finite since X is noetherian. Suppose that $\mathscr{L}_i$ is ample for each $i = 1, \ldots, r$, and consider the finite filtration $\mathscr{F} \supset \mathscr{I}_{X_1}\mathscr{F} \supset \cdots \supset \mathscr{I}_{X_1\cup\cdots\cup X_r}\mathscr{F} = 0$, where each $X_1 \cup \cdots \cup X_k$ is given the reduced induced subscheme structure. We once again receive exact sequences

$$0 \to \mathscr{I}_{Y_1\cup\cdots\cup Y_{k+1}}\mathscr{F} \to \mathscr{I}_{Y_1\cup\cdots\cup Y_k}\mathscr{F} \to \frac{\mathscr{I}_{Y_1\cup\cdots\cup Y_k}\mathscr{F}}{\mathscr{I}_{Y_1\cup\cdots\cup Y_{k+1}}\mathscr{F}} \to 0$$

for all $0 \leqslant k < r$, where

$$\mathscr{I}_{Y_{k+1}} \subset \operatorname{Ann}\left(\frac{\mathscr{I}_{Y_1\cup\cdots\cup Y_k}\mathscr{F}}{\mathscr{I}_{Y_1\cup\cdots\cup Y_{k+1}}\mathscr{F}}\right).$$

As above, we have coherent sheaves $\mathscr{G}_{k+1}$ on Y_{k+1} such that

$$H^\bullet(X, \frac{\mathscr{I}_{Y_1\cup\cdots\cup Y_k}\mathscr{F}}{\mathscr{I}_{Y_1\cup\cdots\cup Y_{k+1}}\mathscr{F}}) \cong H^\bullet(Y_{k+1}, \mathscr{G}_{k+1}).$$

Then (5.3) supplies us with $n_0 > 0$ such that

$$H^p(Y_{k+1}, \mathscr{G}_{k+1} \otimes \mathscr{L}_{k+1}^n) = 0$$

for all $p > 0$ and $n \geqslant n_0$. Once again, this gives us surjectivity of the maps

$$H^p(X, \mathscr{I}_{Y_1\cup\cdots\cup Y_{k+1}}\mathscr{F}) \to H^p(X, \mathscr{I}_{Y_1\cup\cdots\cup Y_k}\mathscr{F})$$

whenever $p > 0$, so the result follows by induction on $r - k$.

(d) Suppose first that $\mathscr{L}$ is ample on X. Let $\mathscr{F}$ be a coherent sheaf on Y. By (II, Ex. 5.1d), we have

$$f_*(\mathscr{F} \otimes_{\mathscr{O}_Y} f^*\mathscr{L}^n) \cong f_*\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^n.$$

X, Y are noetherian and f is a finite map, so $f_*\mathscr{F}$ is coherent by (II, Ex. 5.5). Then (5.3) gives us $n_0 > 0$ such that, for all $n > 0$ and $p > 0$,

$$H^p(Y, \mathscr{F} \otimes f^*\mathscr{L}^n) \cong H^p(X, f_*\mathscr{F} \otimes \mathscr{L}^n) \cong 0$$

by (2.10). By (5.3) once more, we conclude that $f^*\mathscr{L}$ is ample.

Suppose now that $f^*\mathscr{L}$ is ample. □

5.8. Prove that every one-dimensional proper scheme X over an algebraically closed field k is projective.

(a) If X is irreducible and nonsingular, then X is projective by (II, 6.7).

(b) If X is integral, let $\widetilde{X}$ be its normalization (II, Ex. 3.8). Show that $\widetilde{X}$ is complete and nonsingular, hence projective by (a).
Let $f : \widetilde{X} \to X$ be the projection. Let $\mathscr{L}$ be a very ample invertible sheaf on $\widetilde{X}$. Show there is an effective divisor $D = \sum P_i$ on $\widetilde{X}$ with $\mathscr{L}(D) \cong \mathscr{L}$, and such that $f(P_i)$ is a nonsingular point of X, for each i.
Conclude that there is an invertible sheaf $\mathscr{L}_0$ on X with $f^*\mathscr{L}_0 \cong \mathscr{L}$. Then use (Ex. 5.7d), (II, 7.6), and (II, 5.16.1) to show that X is projective.

(c) If X is reduced, but not necessarily irreducible, let $X_1, \ldots, X_r$ be the irreduciblee components of X. Use (Ex. 4.5) to show that $\operatorname{Pic} X \to \bigoplus \operatorname{Pic} X_i$ is surjective. Then use (Ex. 5.7c) to show X is projective.

(d) Finally, if X is any one dimensional proper scheme over k, use (2.7) and (Ex. 4.6) to show that $\operatorname{Pic} X \to \operatorname{Pic} X_{\text{red}}$ is surjective. Then use (Ex. 5.7b) to show X is projective.

Proof. (a) This is exact (ii) $\Longrightarrow$ (i) of (II, 6.7).

(b) □